

VANTARA PETROCHEM - UNIT 3

LUBRICATION STANDARD

Document Number: LUB-STD-001 | Revision: 3 | Effective: 2022-01-15

1. SCOPE AND PURPOSE

This standard defines approved lubricants for rotating equipment at Vantara Petrochem Unit 3. All maintenance personnel shall follow these requirements when selecting lubricants for equipment lubrication, oil changes, and top-ups. Use of non-approved lubricants may void equipment warranties and cause premature component failure.

2. APPROVED LUBRICANTS

Stock Code	Product Name	Type	Approved Applications
LUB-001	VantaLube Premium 320	Mineral Oil (Group II)	All rotating equipment including pumps, compressors, gearbox
LUB-002	SynthMax HP 220	Synthetic PAO	High-temperature applications above 120°C
LUB-003	EconoLube Standard 320	Mineral Oil (Group I)	NON-CRITICAL equipment only - see Section 4 restrictions

3. EQUIPMENT-SPECIFIC REQUIREMENTS

Equipment Type	Seal Type	Required Lubricant	Notes
Centrifugal Pumps (Critical)	FFKM/Fluorocarbon	LUB-001 or LUB-002 ONLY	Includes P-101A, P-101B
Centrifugal Pumps (Non-critical)	NBR/Nitrile	LUB-001, LUB-002, or LUB-003	
Reciprocating Compressors	PTFE Packing	LUB-001 or LUB-002	Includes C-102
Gearboxes	Lip Seals	LUB-001, LUB-002, or LUB-003	

4. LUBRICANT RESTRICTIONS - CRITICAL

WARNING - ECONOLUBE STANDARD 320 (LUB-003) RESTRICTIONS:

EconoLube Standard 320 contains ester-based extreme pressure (EP) additives that are **INCOMPATIBLE** with fluorocarbon elastomers (FFKM, FKM, Viton). Contact between EconoLube Standard 320 and fluorocarbon seal materials causes:

- Elastomer swelling and dimensional changes
- Chemical degradation of seal faces
- Loss of sealing integrity
- Premature mechanical seal failure (typically within 6-10 weeks of exposure)

DO NOT USE LUB-003 on any equipment with fluorocarbon/FFKM seals, including:

- P-101A Primary Feed Pump (John Crane Type 4620 seal with FFKM elastomers)
- P-101B Standby Feed Pump (John Crane Type 4620 seal with FFKM elastomers)
- Any pump handling hydrocarbons above 80°C

Failure to comply with this restriction may result in seal failure, hydrocarbon release, and potential safety incidents.

5. TECHNICAL BASIS

Seal Material Compatibility:

Fluorocarbon elastomers (FFKM/FKM) are susceptible to degradation when exposed to ester-based compounds. The ester additives in Group I mineral oils with EP packages cause the fluorocarbon polymer chains to absorb the ester molecules, resulting in swelling of 15-25% and loss of mechanical properties.

OEM Requirements:

John Crane Technical Bulletin JC-TB-2019-042 specifies that Type 4620 cartridge seals with FFKM elastomers require lubricants free of ester-based additives. Use of non-compatible lubricants voids the seal warranty.

Flowserve Pump Manual Reference:

Flowserve HPX Series Installation Manual Section 7.3.2 states: "Use only lubricants approved for fluorocarbon seal compatibility. Ester-based lubricants will cause premature seal failure."

6. MANAGEMENT OF CHANGE

Per OISD-STD-144 Section 4.5.2, any change to approved lubricants requires:

1. Technical review by Reliability Engineering
2. Compatibility assessment with equipment seal materials
3. MOC documentation and approval
4. Update to this standard

Substitution of lubricants without MOC approval is prohibited.

DOCUMENT APPROVAL

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